

Rare type I collagen variants in early-onset bicuspid aortic valve disease: Overlap with Ehlers-Danlos syndrome and osteogenesis imperfecta

Sara Mansoorshahi,¹ Catherina Tovar Pensa,¹ Erin Carter,² Anna Sabate-Rotes,³ Hector I. Michelena,⁴ Anji T. Yetman,⁵ Julie De Backer,⁶ Laura Muiño Mosquera,⁶ Malenka M. Bissell,⁷ Maria Grazia Andreassi,⁸ Ilenia Foffa,⁸ Dawn S. Hui,⁹ Anthony Caffarelli,¹⁰ Yuli Y. Kim,¹¹ Rodolfo Citro,¹² Margot De Marco,¹³ Justin T. Tretter,¹⁴ Kim L. McBride,¹⁵ Simon C. Body,¹⁶ Cathleen Raggio,² Dianna M. Milewicz,¹ Siddharth K. Prakash,^{1,17,*} and the EBAV Investigators

Summary

Bicuspid aortic valve (BAV) is the most common congenital heart lesion in adults and is often associated with thoracic aortic aneurysms and aortic stenosis. The genetic causes of most non-syndromic cases of BAV remain unknown. Pathogenic variants in *COL1A1* or *COL1A2*, which encode type I collagen (COL1), cause osteogenesis imperfecta (OI), a rare disorder marked by bone fragility. Although aortic valve phenotypes, including BAV, have been described in OI, COL1 genes have not been implicated in sporadic BAV. We report rare COL1 variants in individuals with early-onset BAV (EBAV) complications. Whole-exome sequencing was performed in 272 non-syndromic BAV probands who developed valve or aortic complications before age 30 (EBAV) and 272 biological relatives. Variants were filtered by allele frequency, inheritance pattern, ClinVar classification, and *in silico* predictions. Participants with rare predicted damaging COL1 variants were recontacted to confirm phenotypes. Rare coding variants in *COL1A1* ($n = 5$) and *COL1A2* ($n = 5$) were identified in 10 (4%) EBAV probands, representing a 30-fold enrichment compared with European ancestry populations. Only two variants involved glycine substitutions in triple-helical domains. Affected probands presented with aortic regurgitation and/or thoracic aneurysms requiring repair and exhibited subtle connective tissue features such as joint hypermobility, recurrent fractures, or dental abnormalities. Predicted damaging COL1 variants are enriched in EBAV probands with features overlapping OI and Ehlers-Danlos syndrome, though involving different residues than those in OI. Genetic testing for these variants may help identify individuals who could benefit from personalized surveillance or targeted preventive strategies.

Introduction

Type I collagen (COL1), encoded by *COL1A1* (alpha-1; MIM: 120150) and *COL1A2* (alpha-2; MIM: 120160), is one of the most abundant proteins in the human body and provides mechanical scaffolding for bone, skin, tendons, arteries, and other connective tissues.¹ COL1 is synthesized as a heterotrimer of two alpha-1 chains and one alpha-2 chain. It is highly expressed in the aortic media and adventitia, where it contributes tensile strength to the aorta.² Pathogenic variants (PVs) in *COL1A1* or *COL1A2* cause osteogenesis imperfecta (OI) types I–IV, four subtypes of Ehlers-Danlos syndrome (EDS), and COL1-related overlap disorder (C1ROD; MIM: 618172) with mixed variable features of OI and EDS.

OI refers to a group of related heritable disorders characterized by bone fragility. The population prevalence of OI is 6 per 100,000 but with substantial clinical and genetic heterogeneity.^{3,4} OI type I (MIM: 166200) is the most common subtype and the mildest form of OI.¹ Individuals with type I OI typically present due to fractures after minor trauma, with other variable findings such as blue sclerae, hearing loss, and brittle teeth.³ Types II (MIM: 166210) and III (MIM: 259420) are considered to be the most severe forms of OI due to spontaneous neonatal fractures with substantial deformities or perinatal death.⁵ Type IV OI (MIM: 166220) is characterized by an intermediate phenotype of mild to moderate severity. While OI primarily affects the musculoskeletal system, valvular heart disease, myocardial dysfunction, atrial fibrillation,

¹Department of Internal Medicine, University of Texas Health Science Center at Houston, Houston, TX, USA; ²Hospital for Special Surgery, 535 East 70th Street, New York, NY, USA; ³Department of Pediatric Cardiology, Vall d'Hebron University Hospital, Barcelona, Spain; ⁴Department of Cardiovascular Medicine, Mayo Clinic, Rochester, MN, USA; ⁵Children's Hospital and Medical Center, University of Nebraska, Omaha, NE, USA; ⁶Center for Medical Genetics, Ghent University, Ghent, Belgium; ⁷Department of Biomedical Imaging Science, Leeds Institute of Cardiovascular and Metabolic Medicine, University of Leeds, Leeds, UK; ⁸Consiglio Nazionale delle Ricerche (CNR), L'Istituto di Fisiologia Clinica, Pisa, Italy; ⁹Department of Cardiothoracic Surgery, University of Texas Health Science Center San Antonio, San Antonio, TX, USA; ¹⁰Hoag Hospital Newport Beach, Newport Beach, CA, USA; ¹¹Division of Cardiovascular Medicine, Perelman School of Medicine at the University of Pennsylvania, Philadelphia, PA, USA; ¹²Medicine and Health Science "V. Tiberio" Department, University of Molise Responsible Research Hospital, Campobasso, Italy; ¹³Department of Medicine, Surgery and Dentistry Schola Medica Salernitana, University of Salerno, Baronissi, Italy; ¹⁴Department of Pediatric Cardiology, Cleveland Clinic, Cleveland, OH, USA; ¹⁵Department of Medical Genetics, Cumming School of Medicine, University of Calgary, Calgary, AB, Canada; ¹⁶Department of Anesthesiology, Boston University School of Medicine, Boston, MA, USA

¹⁷Lead contact

*Correspondence: siddharth.k.prakash@uth.tmc.edu

<https://doi.org/10.1016/j.xhgg.2026.100627>.

© 2026 The Author(s). Published by Elsevier Inc. on behalf of American Society of Human Genetics.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).



and arterial hypertension appear to be increased in OI cohorts.^{6,7} The prevalence of thoracic aortic aneurysms (TAAs) and ruptured cerebral aneurysms may also be increased in OI. A recent literature review concluded that a low threshold is reasonable for the investigation of suspected cardiovascular disease in OI but that there is currently insufficient evidence to recommend routine screening of individuals with OI for cardiac or vascular abnormalities.⁸

The EDS subtypes are a heterogeneous group of more than 10 conditions that affect the integrity of connective tissues.⁹ Common elements of the EDS phenotype include joint hypermobility, striae, and abnormal wound healing.^{10,11} The classical (cEDS; MIM: 130000), arthrochalasia (aEDS; MIM: 130060), vascular (vEDS; MIM: 130050), and cardiac valvular (MIM: 225320) subtypes of EDS are associated with PV in type I, III, or V collagen.^{9,11} Cardiovascular complications of collagen-related EDS subtypes are frequent and include arterial aneurysms, spontaneous arterial dissection or rupture, and valvular heart disease.¹² Arterial dissections appear to be even more frequent in C1ROD than in OI or EDS.¹³ Therefore, surveillance of some individuals with collagen PV using echocardiography or angiography may be reasonable to exclude cardiovascular lesions.

Bicuspid aortic valve (BAV; MIM: 109730) is the most common congenital heart malformation (1%) and is best characterized by autosomal-dominant inheritance of rare variants in many different genes with incomplete penetrance and highly variable phenotypic expressivity. There is strong cumulative evidence that *GATA4*, *GATA6*, *NOTCH1*, *ROBO4*, *SMAD4*, *MUC4*, and *SMAD6* each contribute to a small percentage of non-syndromic BAV cases.¹⁴ BAV is associated with an increased lifetime risk for TAAs (MIM: 132900), predisposing to aortic dissections and valvular heart disease requiring surgical repair. BAV-related complications usually increase after age 40, but rare individuals experience significant complications as children or young adults. We hypothesize that young adults with early-onset complications of BAV disease (early-onset BAV [EBAV]) represent a phenotypic extreme enriched for rare genetic variants with large biological effects. Consistent with this framework, we identified recurrent rare variants of cardiac developmental genes, including *COL1A1* and *COL1A2*, in a cohort of individuals selected based on EBAV phenotypes.¹⁵ Collagen variants had not been previously implicated in familial BAV studies or in cohorts dominated by older individuals with later-onset disease.¹⁶ We therefore hypothesize that *COL1* variants may act as genetic modifiers to accelerate the onset or severity of BAV-associated valve and aortic disease.

Subjects and methods

The Committee for the Protection of Human Subjects at the University of Texas Health Science Center at Houston approved

the study protocol (HSC-MS-11-0185), and all participants provided informed consent prior to enrollment. The Genetic Risks for Bicuspid Aortic Valve Disease study ([ClinicalTrials.gov: NCT01823432](https://clinicaltrials.gov/ct2/show/study/NCT01823432)) was an international registry that collected samples and clinical data from probands with heritable non-syndromic BAV disease, as well as first-degree relatives and affected family members. Participants were enrolled at expert centers where they received clinical care in accordance with the 2022 ACC/AHA Guideline for the Diagnosis and Management of Aortic Disease.¹⁷

We defined early-onset complications of BAV (EBAV) as clinical presentation due to moderate or severe aortic valve stenosis or regurgitation or a large TAA (Z score > 4), or aortic valve or aortic surgery, aortic dissection, or intervention for other congenital left-sided lesions prior to age 30.^{15,18} Prospective adult BAV cohorts are predominantly middle-aged with the first presentation in the fourth to sixth decade of life, reflecting the typical timing of clinically apparent valvulo-aortic disease in adulthood. In a large population-based cohort, complex valvulo-aortopathy, defined similarly to EBAV, accounted for only ~7% of adult BAV diagnoses, supporting the view that EBAV-like phenotypes are relatively uncommon in adults.¹⁹ Aortic valve stenosis or regurgitation was defined according to AHA/ACC guidelines.²⁰ Individuals with known PVs, recognized genetic syndromes, non-recurrent neonatal coarctation, or complex congenital heart disease were excluded.

Whole-exome sequencing (WES) of EBAV probands, plus parents, siblings, and any other biological relatives with BAV, was performed at the University of Washington Center for Rare Disease Research (UW-CRDR) as previously reported.¹⁵ Variant filtration was performed using a customized version of *seqr* at the UW-CRDR (v.1.0-4cb7dfc3) with the following settings: *de novo* or dominant inheritance model; pathogenic, likely pathogenic, variants of uncertain significance [VUSs], or conflicting ClinVar classifications; nonsense, essential splice-site, in-frame insertion or deletion (indel), frameshift, or missense variants; allele frequency <0.0001; REVEL >0.5; CADD >20; genotype quality >20; and allele balance >0.25. Predicted damaging missense or in-frame indel (pDMI) variants were retained if they were predicted to be damaging by at least 3 of PolyPhen, MutTaster, SIFT, or FATHMM.

We recontacted EBAV study participants with rare *COL1* variants to complete surveys self-reporting clinical characteristics related to OI and EDS. The assessed phenotypic features were bone fragility, blue sclerae, long bone deformities, and dentinogenesis imperfecta,³ defined as opalescent teeth and/or a history of fractured teeth, as well as skin distensibility, atrophic scarring, joint hypermobility, frequent bruising, and hernias.¹¹ We also screened medical records for evidence of phenotypes related to OI or EDS.

We compared the phenotypes and genotypes of individuals with EBAV and rare *COL1* variants to a cohort of adults with types I, III, and IV OI who were recruited into an institutional review board (IRB)-approved study of cardiopulmonary manifestations during routine clinic visits between August 2017 and May 2019 at the Kathryn O. and Alan C. Greenberg Center for Skeletal Dysplasias at the Hospital for Special Surgery and to a cohort of adults with heritable thoracic aortic aneurysms and dissections (HTAD).²¹ The HTAD dataset consists of whole-exome and whole-genome sequences of unrelated probands with two or more biological relatives affected by thoracic aortic disease or a history of aneurysm surgery or dissection at ≤60 years of age

Table 1. Missense COL1 variants of EBAV and OI probands

Gene	Cohort	POS	NT	cDNA	aa	Ann	No.	CADD	REVEL	PP	Structural impact
COL1A1	EBAV	50189879	G>A	c.2593C>T (GenBank: NM_000088.4)	p.Arg865Cys (GenBank: NP_000079.2)	V	10	32	0.875	0.855	introduces a cysteine into Y-position 687 of the triple-helical domain
COL1A1	EBAV	50189173	G>A	c.2932C>T (GenBank: NM_000088.4)	p.Pro978Ser (GenBank: NP_000079.2)	V	469	24	0.71	0.996	deletes a proline at Y-position 800 in the triple-helical domain that is typically hydroxylated
COL1A1	EBAV	50188764	C>T	c.3077G>A (GenBank: NM_000088.4)	p.Arg1026Gln (GenBank: NP_000079.2)	V	4	26	0.752	0.979	change in charge at X-position 844 in the triple-helical domain
COL1A1	EBAV	50187967	C>T	c.3278G>A (GenBank: NM_000088.4)	p.Arg1093His (GenBank: NP_000079.2)	V	97	32	0.734	0.855	relatively conservative substitution at Y-position 915 of the Gly-Xaa-Yaa motif in the triple-helical domain
COL1A1	EBAV	50186688	C>T	c.3766G>A (GenBank: NM_000088.4)	p.Ala1256Thr (GenBank: NP_000079.2)	V	22	25	0.91	0.359	relatively conservative substitution in a highly conserved region of the C-propeptide
COL1A1	OI	50197045	C>T	c.769G>A (GenBank: NM_000088.4)	p.Gly257Arg (GenBank: NP_000079.2)	P	1	32	0.996	1	Gly substitution in the Gly-Xaa-Yaa motif at position 79 of the triple-helical domain
COL1A1	OI	50190099	C>T	c.2461G>A (GenBank: NM_000088.4)	p.Gly821Ser (GenBank: NP_000079.2)	P	0	28	0.985	0.974	Gly substitution in the Gly-Xaa-Yaa motif at position 634 of the triple-helical domain
COL1A1	OI	50188601	C>A	c.3136G>T (GenBank: NM_000088.4)	p.Gly1046Cys (GenBank: NP_000079.2)	P	0	27	0.961	1	Gly substitution in the Gly-Xaa-Yaa motif at position 868 of the triple-helical domain
COL1A2	EBAV	94409359	G>C	c.830G>C (GenBank: NM_000089.4)	p.Gly277Ala (GenBank: NP_000080.2)	P	59	25	0.889	0.705	Gly substitution in the Gly-Xaa-Yaa motif at position 187 of the triple-helical domain
COL1A2	EBAV	94420275	C>T	c.2122C>T (GenBank: NM_000089.4)	p.Arg708Trp (GenBank: NP_000080.2)	V	7	32	0.882	0.998	non-conservative substitution at X-position 530 of the triple-helical domain
COL1A2	EBAV	94427672	G>A	c.3313G>A (GenBank: NM_000089.4)	p.Gly1105Ser (GenBank: NP_000080.2)	V	1,438	24	0.783	1	first residue of the telopeptide after the triple-helical domain that is retained after C-propeptide cleavage
COL1A2	EBAV	94427849	C>T	c.3490C>T (GenBank: NM_000089.4)	p.Arg1164Cys (GenBank: NP_000080.2)	V	10	28	0.801	0.999	adjacent to first cysteine of the C-propeptide that forms a loop required for heterotrimer formation
COL1A2	EBAV	94427852	G>A	c.3493G>A (GenBank: NM_000089.4)	p.Asp1165Asn (GenBank: NP_000080.2)	V	4	32	0.695	0.999	in 6-residue calcium-binding domain of the C-propeptide
COL1A2	OI	94410278	G>A	c.1072G>A (GenBank: NM_000089.4)	p.Gly358Ser (GenBank: NP_000080.2)	P	1	32	0.981	0.999	Gly substitution in the Gly-Xaa-Yaa motif at position 180 of the triple-helical domain
COL1A2	OI	94410899	G>T	c.1208G>T (GenBank: NM_000089.4)	p.Gly403Val (GenBank: NP_000080.2)	P	0	27	0.96	0.924	Gly substitution in the Gly-Xaa-Yaa motif at position 225 of the triple-helical domain

(Continued on next page)

Table 1. Continued

Gene	Cohort	POS	NT	cDNA	aa	Ann	No.	CADD	REVEL	PP	Structural impact
<i>COL1A2</i>	OI	94412620	G>C	c.1441G>C (GenBank: NM_000089.4)	p.Gly481Arg (GenBank: NP_000080.2)	P	0	27	0.984	0.974	Gly substitution in the Gly-Xaa-Yaa motif at position 303 of the triple-helical domain
<i>COL1A2</i>	OI	94417734	G>C	c.1874G>C (GenBank: NM_000089.4)	p.Gly625Ala (GenBank: NP_000080.2)	LP	1	28	0.988	0.999	Gly substitution in the Gly-Xaa-Yaa motif at position 447 of the triple-helical domain

Loss-of-function COL1 variants were not included. EBAV, early-onset BAV disease; OI, type I osteogenesis imperfecta; POS, GrCh38 genomic position; NT, nucleotide change; aa, amino acid change; C, nucleotide-level descriptor (HGVS); P, protein-level descriptor (HGVS); ANN, annotation of variant (P, pathogenic; LP, likely pathogenic; V, variant of uncertain significance); No., number of gnomAD 4.1 alleles in non-Finnish Europeans; PP, PolyPhen score.

without affected relatives who were recruited by the John Ritter Research Program in Aortic and Vascular Diseases at the University of Texas Health Science Center at Houston.²²

Results

Characteristics of study cohorts

The EBAV cohort included 544 individuals (98% of European ancestry): 272 EBAV probands, 21 relatives with BAV, and 251 apparently unaffected family members, including 26 trios and 15 multiplex families. All EBAV probands were evaluated clinically to exclude genetic syndromes. Detailed phenotypic features of this cohort were previously reported.²³ The average age at diagnosis was 18 years, with prevalent co-existing non-complex congenital heart lesions and recurrent aortic interventions.

The OI cohort consisted of 34 individuals (40 [20–70] years, 80% women): 20 with *COL1A1* variants, 7 with *COL1A2* variants, 2 with variants in non-COL1 genes, and 5 without genetic test results. Four individuals (13%) had a history of cardiovascular disease: hypertension, TAAs, coronary artery disease, and/or stroke. Eighteen of the 23 participants who were imaged had valve or structural heart abnormalities such as TAAs involving the proximal aorta, atrial septal defects, and mild or greater valvular regurgitation. None of the individuals in the OI cohort was reported to have a BAV. Family members were not available for segregation analysis.

Characteristics of rare COL1 variants

Predicted damaging rare *COL1A1* or *COL1A2* missense variants were identified in 10 EBAV probands (4%, Table 1). This represents a 30-fold enrichment relative to European ancestry gnomAD control subjects. While EBAV variants were distributed throughout the coding regions of *COL1A1* and *COL1A2*, 8 of 10 were located in the distal half of the triple-helical domains or C-terminal propeptides (Figure 1). Most of the COL1 EBAV variants are classified as VUSs and are associated with OI and/or EDS phenotypes in ClinVar (Table 2).

We compared the EBAV-associated COL1 variants to COL1 missense PV in adults with types I, II, and IV OI,

C1ROD, and HTAD. While all OI missense variants were glycine substitutions in the triple-helical domains, only two EBAV-associated variants were glycine substitutions affecting *COL1A2*, and only one EBAV-associated variant, c.830G>C (GenBank: NM_000089.4) (p.Gly277Ala), was located in the triple-helical domain and is classified as pathogenic in ClinVar. Notably, this variant was also observed in one HTAD family with two affected individuals who have the Gly277Ala variant and have BAV. The other glycine substitution, c.3313G>A (GenBank: NM_000089.4) (p.Gly1105Ser), is classified as a VUS. Loss-of-function variants affecting *COL1A1* (56%) and *COL1A2* (10%) were frequent in the OI cohort but absent in the EBAV or HTAD cohort. C1ROD variants clustered in the N-terminal region of the *COL1A1* triple-helical domain. In contrast, EBAV and HTAD variants clustered in the C-terminal triple-helical and C-propeptide domains of *COL1A1* and the N- and C-terminal regions of the *COL1A2* triple-helical domain.

Phenotypes of individuals with rare COL1 variants

We successfully recontacted 6 of the 10 EBAV participants with rare COL1 variants. All were diagnosed with moderate or severe aortic valve stenosis and/or aortic valve regurgitation or large ($Z > 4$) TAAs (Table 2). While none of these individuals had been diagnosed with OI, two reported significant features that may be commonly observed in type I OI. One EBAV proband, harboring *COL1A1* c.3278G>A (p.Arg1093His) reported a history of fractures after minor trauma. Another EBAV proband with *COL1A2* c.3313G>A (p.Gly1105Ser) reported moderate long bone deformities (bowing) and scoliosis that did not require intervention, as well as dentinogenesis imperfecta.

Four EBAV participants who did not have a clinical diagnosis of cEDS reported significant features that are associated with cEDS. One participant, who harbored *COL1A2* c.3493G>A (p.Asp1165Asn) was clinically diagnosed with cEDS and reported hypermobility-related complications that led to corrective joint surgeries. Another proband, with a *COL1A1* c.3278G>A (p.Arg1093His) variant, reported atrophic scarring and skin distensibility. Two other probands, harboring *COL1A2* c.3490C>T (p.Arg1164Cys) and

Table 2. Phenotypes of EBAV participants with rare COL1 variants

Gene	aa	Age at enrollment	Aortic phenotypes	Aortic operations	OI	cEDS	ClinVar OI	ClinVar EDS
<i>COL1A1</i>	p.Arg865Cys	38	AS, TAA	AVR, PAR	0	0	0	0
<i>COL1A1</i>	p.Pro978Ser	38	AS, AR, TAA	AVR, PAR	N/A	N/A	2	2
<i>COL1A1</i>	p.Arg1026Gln	29	AS, AR	none	0	0	1	0
<i>COL1A1</i>	p.Arg1093His	29	AR	none	1	1	2	1
<i>COL1A1</i>	p.Ala1256Thr	50	NA	N/A	N/A	N/A	2	2
<i>COL1A2</i>	p.Gly277Ala	32	AR, TAA	AVR, PAR	N/A	N/A	1	1
<i>COL1A2</i>	p.Arg708Trp	15	AS	AVR	N/A	N/A	1	1
<i>COL1A2</i>	p.Gly1105Ser	17	AS, AR	none	1	1	2	3
<i>COL1A2</i>	p.Arg1164Cys	29	AR	Coarct, VSD	0	1	1	1
<i>COL1A2</i>	p.Asp1165Asn	17	AR	none	0	1	2	2

We were unable to recontact or obtain records for the individual with the p.Ala1256Thr variant. aa, amino acid change; AVR, aortic valve repair or replacement; PAR, proximal aortic replacement; AS, aortic stenosis; AR, aortic regurgitation; TAA, thoracic aortic aneurysm; Coarct, coarctation of the aorta; VSD, ventricular septal defect; OI, osteogenesis imperfecta phenotypes (bone fragility, blue sclerae, bone deformities, and dentinogenesis imperfecta); cEDS, classical Ehlers-Danlos phenotypes (skin hyperextensibility, atrophic scars, joint hypermobility, easy bruising, and hernias); N/A, not available; ClinVar OI, number of individuals in ClinVar carrying this variant who are reported with OI features; ClinVar EDS, number of individuals in ClinVar carrying this variant who are reported with EDS features.

the N-protease cleavage site. Three of the EBAV rare variants affect conserved residues in the procollagen C-propeptide domains of *COL1A1* and *COL1A2*. PVs affecting the C-propeptides can disrupt procollagen trimerization or collagen fibril formation.³⁰ These abnormalities in COL1 assembly or structure are predicted to decrease collagen maturation, fibril assembly, and tissue mechanical integrity, which would, in turn, lead to altered extracellular matrix properties and reduced tensile strength of the aortic valve and aortic wall, potentially accelerating valvular and aortic disease. We hypothesize that EBAV COL1 variants may cause a distinct overlap phenotype with less prominent OI phenotypes and more prominent cardiovascular disease phenotypes, including TAA and BAV.

As in the general population, cardiovascular disease is the leading cause of death for individuals with OI. Thoracic aortic dilation and mitral or aortic valve regurgitation appear to be increased in OI, but the clinical implications of cardiac findings in OI are currently unclear due to missing data and conflicting observations.⁸ Notably, life-threatening cardiovascular complications, including arterial dissections, were reported in two small C1ROD cohorts and may be increased in comparison to OI phenotypes, which are considered to be milder than OI caused by glycine substitutions in the C-terminal regions of the triple-helical domains.^{25,27,28} Precise cardiovascular phenotyping of OI and C1ROD cohorts is needed to confirm the prevalence of BAV and arterial aneurysms or dissections and to identify genotype-phenotype correlations between COL1 variants and cardiovascular lesions.

COL1A1 or *COL1A2* variants were not found to be directly associated with BAV in several genome-wide association studies (GWASs) of older participants who predominantly presented due to aortic stenosis.^{16,31,32} The phenotype of the EBAV cohort is distinct from the

GWAS cohorts due to youth (20 years at initial presentation) and valve function (predominant aortic regurgitation). In this context, COL1 variants most likely function as modifier alleles that accelerate disease onset or severity. The milder predicted effects and gnomAD allele counts of most EBAV-associated variants are consistent with putative roles as risk modifiers rather than causative alleles. Disruption of the mechanical resilience of valve or aortic tissues due to rare COL1 variants could exacerbate disease progression in individuals with BAV or TAA by increasing susceptibility to cytotoxic or hemodynamic stress from smoking and hypertension, promoting earlier or more severe manifestations of BAV disease. Common and rare variants may influence BAV risk through shared extracellular matrix pathways: three of the top seven genes identified in BAV GWASs (*PALMD*, *TGFBI*, and *ADAMTSL1*) have been linked to altered COL1 deposition or organization in valve or cardiac tissues.^{33–35}

There was insufficient phenotypic data on EBAV families to perform detailed correlations between COL1 genotypes and specific clinical features, valve configuration, or cusp characteristics. It was not possible to determine if rare COL1 variants segregated with BAV in all families due to incomplete genotypic data. Some contributory genetic variants may have been filtered out due to the stringency of the selection criteria. This study was restricted to adults, and additional studies are needed to determine if COL1 variants contribute to pediatric-onset BAV disease. The EBAV cohort was largely of European descent, and therefore, the results of this study may not be generalizable to non-European cohorts.

The findings of this study implicate *COL1A1* and *COL1A2* as significant risk genes for EBAV disease, but additional studies are needed to verify that COL1 rare variants modify the severity of BAV-related complications. If these observations are replicated, careful assessment of

EDS or OI phenotypes and/or genotyping of COL1 variants may eventually prove useful for clinical risk stratification of individuals with BAV who have early-onset features to guide management or prevent future complications.

Data and code availability

Sequence data were deposited in NHGRI's Analysis Visualization and Informatics Lab-space (AnVIL) and are available through dbGaP accessions phs000693 (University of Washington Center for Mendelian Genomics) and phs003047 (UW-CRDR).

Consortia

The EBAV Investigators are Shaine A. Morris, Rita Milewski, Giuseppe Limongelli, Allesandro Della Corte, Laura Perone, Yuli Y. Kim, Hector Michelena, Maria G. Andreassi, Arturo Evangelista, Denver Saltee, Anji Yetman, Kim McBride, Eduardo Bossone, Rodolfo Citro, Dawn S. Hui, Malenka M. Bissell, Andrea Ballotti, Ilenia Foffa, Margot De Marco, Anthony Caffarelli, Rita Weise, Julie DeBacker, Laura Muino Mosquera, Robbin Cohen, Laura Dos Subira, Justin T. Tretter, Anna Sabe Rotes, Martina Caiazza, Lamia Ait Ali, Francesca Pluchinotta, and Simon C. Body.

Acknowledgments

We thank Joana Castillo and Jacqueline Jennings for sample preparation, and Abraham Arenas assisted with data analysis. This study was supported in part by R01HL137028 (S.K.P.), R21HL150383 (S.K.P.), R01HL114823 (S.C.B.), and R21HL150373 (S.C.B.) from the National Institutes of Health. Sequencing and data analysis were provided by the University of Washington Center for Rare Disease Research (UW-CRDR), with support from NHGRI grants U01 HG011744 and U24 HG011746. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Declaration of interests

The authors declare no competing interests.

Web resources

ClinicalTrials.gov, <https://clinicaltrials.gov>
dbGaP, <https://dbgap.ncbi.nlm.nih.gov/home/>
GenBank, <https://www.ncbi.nlm.nih.gov/genbank>
OMIM, <https://www.omim.org>

Received: November 7, 2025

Accepted: May 12, 2026

References

- Lu, Y., Zhang, S., Wang, Y., Ren, X., and Han, J. (2019). Molecular mechanisms and clinical manifestations of rare genetic disorders associated with type I collagen. *Intractable Rare Dis. Res.* 8, 98–107. <https://doi.org/10.5582/irdr.2019.01064>.
- Vouyouka, A.G., Pfeiffer, B.J., Liem, T.K., Taylor, T.A., Mudaliar, J., and Phillips, C.L. (2001). The role of type I collagen in aortic wall strength with a homotrimeric $\alpha 1(I)$ 3 collagen mouse model. *J. Vasc. Surg.* 33, 1263–1270. <https://doi.org/10.1067/mva.2001.113579>.
- Steiner, R.D., and Basel, D. (2005). COL1A1/2 Osteogenesis Imperfecta. In *GeneReviews®* [Internet], M.P. Adam, J. Feldman, G.M. Mirzaa, R.A. Pagon, S.E. Wallace, and A. Amemiya, eds. (University of Washington, Seattle). <http://www.ncbi.nlm.nih.gov/books/NBK1295/>.
- Forlino, A., and Marini, J.C. (2016). Osteogenesis imperfecta. *Lancet* 387, 1657–1671. [https://doi.org/10.1016/S0140-6736\(15\)00728-X](https://doi.org/10.1016/S0140-6736(15)00728-X).
- Elhaddadi, H., Ayyad, A., Messaoudi, S., and Amrani, R. (2024). Osteogenesis Imperfecta Type II: The Lethal Newborn Form Diagnosed in the Postnatal Period. *Cureus* 16, e60945. <https://doi.org/10.7759/cureus.60945>.
- Verdonk, S.J.E., Storoni, S., Micha, D., van den Aardweg, J.G., Versacci, P., Celli, L., de Vries, R., Zhytnik, L., Kamp, O., Bugiani, M., and Eekhoff, E.M.W. (2024). Is Osteogenesis Imperfecta Associated with Cardiovascular Abnormalities? A Systematic Review of the Literature. *Calcif. Tissue Int.* 114, 210–221. <https://doi.org/10.1007/s00223-023-01171-3>.
- Folkestad, L., Hald, J.D., Gram, J., Langdahl, B.L., Hermann, A.P., Diederichsen, A.C., Abrahamsen, B., and Brixen, K. (2016). Cardiovascular disease in patients with osteogenesis imperfecta - a nationwide, register-based cohort study. *Int. J. Cardiol.* 225, 250–257. <https://doi.org/10.1016/j.ijcard.2016.09.107>.
- Folkestad, L., Prakash, S.K., Nagamani, S.C.S., Andersen, N.H., Carter, E., Hald, J.D., Johnson, R.J., Langdahl, B., Peretto, E.M., Raggio, C., et al. (2025). Cardiovascular disease in adults with osteogenesis imperfecta: clinical characteristics, care recommendations, and research priorities identified using a modified Delphi technique. *J. Bone Miner. Res.* 40, 211–221. <https://doi.org/10.1093/jbmr/zjae197>.
- Malfait, F., Francomano, C., Byers, P., Belmont, J., Berglund, B., Black, J., Bloom, L., Bowen, J.M., Brady, A.F., Burrows, N.P., et al. (2017). The 2017 international classification of the Ehlers–Danlos syndromes. *American J. of Med. Genetics Pt. C.* 175, 8–26. <https://doi.org/10.1002/ajmg.c.31552>.
- Miklovic, T., and Sieg, V.C. (2025). Ehlers-Danlos Syndrome. *StatPearls* [Internet] (StatPearls Publishing).
- Malfait, F., Symoens, S., and Syx, D. (1993). Classic Ehlers-Danlos Syndrome. In *GeneReviews®* [Internet], M.P. Adam, J. Feldman, G.M. Mirzaa, R.A. Pagon, S.E. Wallace, and A. Amemiya, eds. (University of Washington, Seattle). <http://www.ncbi.nlm.nih.gov/books/NBK1244/>.
- Malfait, F., Wenstrup, R.J., and De Paepe, A. (2010). Clinical and genetic aspects of Ehlers-Danlos syndrome, classic type. *Genet. Med.* 12, 597–605. <https://doi.org/10.1097/GIM.0b013e3181eed412>.
- Morlino, S., Micale, L., Ritelli, M., Rohrbach, M., Zoppi, N., Vandersteen, A., Mackay, S., Agolini, E., Cociadiferro, D., Sasaki, E., et al. (2020). COL1-related overlap disorder: A novel connective tissue disorder incorporating the osteogenesis imperfecta/Ehlers-Danlos syndrome overlap. *Clin. Genet.* 97, 396–406. <https://doi.org/10.1111/cge.13683>.
- Bray, J.J.H., Freer, R., Pitcher, A., and Kharbanda, R. (2023). Family screening for bicuspid aortic valve and aortic dilatation: a meta-analysis. *Eur. Heart J.* 44, 3152–3164. <https://doi.org/10.1093/eurheartj/ehad320>.

15. Mansoorshahi, S., Yetman, A.T., Bissell, M.M., Kim, Y.Y., Michelena, H.I., De Backer, J., Mosquera, L.M., Hui, D.S., Caffarelli, A., Andreassi, M.G., et al. (2024). Whole-exome sequencing uncovers the genetic complexity of bicuspid aortic valve in families with early-onset complications. *Am. J. Hum. Genet.* *111*, 2219–2231. <https://doi.org/10.1016/j.ajhg.2024.08.001>.
16. Bermudez, F., Shakt, G., Bowles, A., Alba, P., Dinatale, T., Chang, K.M., Tsao, P.S., Lynch, J., Dochtermann, D., Pyarajan, S., et al. (2026). Multipopulation Genome-Wide Association Study Identifies Novel Loci for Bicuspid Aortic Valve and Reveals Shared Genetic Architecture With Aortopathies. *Circ. Genom. Precis. Med.* *19*, e005491. <https://doi.org/10.1161/CIRCGEN.125.005491>.
17. Isselbacher, E.M., Preventza, O., Hamilton Black, J., Augoustides, J.G., Beck, A.W., Bolen, M.A., Braverman, A.C., Bray, B.E., Brown-Zimmerman, M.M., Chen, E.P., et al. (2022). 2022 ACC/AHA Guideline for the Diagnosis and Management of Aortic Disease: A Report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation* *146*, e334–e482. <https://doi.org/10.1161/CIR.0000000000001106>.
18. Campens, L., Demulier, L., De Groote, K., Vandekerckhove, K., De Wolf, D., Roman, M.J., Devereux, R.B., De Paepe, A., and De Backer, J. (2014). Reference values for echocardiographic assessment of the diameter of the aortic root and ascending aorta spanning all age categories. *Am. J. Cardiol.* *114*, 914–920. <https://doi.org/10.1016/j.amjcard.2014.06.024>.
19. Yang, L.T., Ye, Z., Wajih Ullah, M., Maleszewski, J.J., Scott, C.G., Padang, R., Pislaru, S.V., Nkomo, V.T., Mankad, S.V., Pellikka, P.A., et al. (2023). Bicuspid aortic valve: long-term morbidity and mortality. *Eur. Heart J.* *44*, 4549–4562. <https://doi.org/10.1093/eurheartj/ehad477>.
20. Otto, C.M., Nishimura, R.A., Bonow, R.O., Carabello, B.A., Erwin, J.P., Gentile, F., Jneid, H., Krieger, E.V., Mack, M., McLeod, C., et al. (2020). ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation* *143*, e72–e227. <https://doi.org/10.1161/CIR.0000000000000923>.
21. Khan, S.I., Yonko, E.A., Carter, E.M., Dyer, D., Sandhaus, R.A., and Raggio, C.L. (2020). Cardiopulmonary Status in Adults with Osteogenesis Imperfecta: Intrinsic Lung Disease May Contribute More Than Scoliosis. *Clin. Orthop. Relat. Res.* *478*, 2833–2843. <https://doi.org/10.1097/CORR.0000000000001400>.
22. Murdock, D.R., Guo, D.-C., DePaolo, J.S., Schwarze, U., Duan, X.-Y., Cecchi, A.C., Marin, I.C., Tang, Y., Chong, J.X., Bamshad, M.J., et al. (2025). Non-canonical splice variants in thoracic aortic dissection cases and Marfan syndrome with negative genetic testing. *npj Genom. Med.* *10*, 25. <https://doi.org/10.1038/s41525-025-00472-w>.
23. Selbst, M.A., Ward, C.R., Svetgoff, M.L., Michelena, H.I., Sabate-Rotes, A., De Backer, J., Mosquera, L.M., Yetman, A.T., Bissell, M.M., Andreassi, M.G., et al. (2024). Bicuspid Aortic Valve Disease with Early-Onset Complications: Characteristics and Aortic Outcomes. *Hearts (Basel)*. *5*, 253–258. <https://doi.org/10.3390/hearts5030018>.
24. DiGregorio, H., Mansoorshahi, S., Carlisle, S.G., Tovar Pensa, C., Watts, A., McNeely, C., Sabate-Rotes, A., Yetman, A., Michelena, H.I., De Backer, J.F.A., et al. (2025). Contribution of rare chromosome 22q11.2 copy number variants to non-syndromic bicuspid aortic valve. *Heart* *111*, 221–229. <https://doi.org/10.1136/heartjnl-2024-324669>.
25. Gnoli, M., Brizola, E., Tremosini, M., Pedrini, E., Maioli, M., Mosca, M., Bassotti, A., Castronovo, P., Giunta, C., and Sangiorgi, L. (2021). COL1-Related Disorders: Case Report and Review of Overlapping Syndromes. *Front. Genet.* *12*, 640558. <https://doi.org/10.3389/fgene.2021.640558>.
26. Guarnieri, V., Morlino, S., Di Stolfo, G., Mastroianno, S., Mazza, T., and Castori, M. (2019). Cardiac valvular Ehlers-Danlos syndrome is a well-defined condition due to recessive null variants in COL1A2. *Am. J. Med. Genet.* *179*, 846–851. <https://doi.org/10.1002/ajmg.a.61100>.
27. Takeda, R., Yamaguchi, T., Hayashi, S., Sano, S., Kawame, H., Kanki, S., Taketani, T., Yoshimura, H., Nakamura, Y., and Kosho, T. (2022). Clinical and molecular features of patients with COL1-related disorders: Implications for the wider spectrum and the risk of vascular complications. *Am. J. Med. Genet.* *188*, 2560–2575. <https://doi.org/10.1002/ajmg.a.62887>.
28. Venable, E., Knight, D.R.T., Thoreson, E.K., and Baudhuin, L.M. (2023). COL1A1 and COL1A2 variants in Ehlers-Danlos syndrome phenotypes and COL1-related overlap disorder. *American J. of Med. Genetics Pt. C.* *193*, 147–159. <https://doi.org/10.1002/ajmg.c.32038>.
29. Cabral, W.A., Makareeva, E., Colige, A., Letocha, A.D., Ty, J.M., Yeowell, H.N., Pals, G., Leikin, S., and Marini, J.C. (2005). Mutations near amino end of alpha1(I) collagen cause combined osteogenesis imperfecta/Ehlers-Danlos syndrome by interference with N-propeptide processing. *J. Biol. Chem.* *280*, 19259–19269. <https://doi.org/10.1074/jbc.M414698200>.
30. Bourhis, J.-M., Mariano, N., Zhao, Y., Harlos, K., Exposito, J.-Y., Jones, E.Y., Moali, C., Aghajari, N., and Hulmes, D.J.S. (2012). Structural basis of fibrillar collagen trimerization and related genetic disorders. *Nat. Struct. Mol. Biol.* *19*, 1031–1036. <https://doi.org/10.1038/nsmb.2389>.
31. Gehlen, J., Stundl, A., Debiec, R., Fontana, F., Krane, M., Sharipova, D., Nelson, C.P., Al-Kassou, B., Giel, A.S., Sinning, J.M., et al. (2023). Elucidation of the genetic causes of bicuspid aortic valve disease. *Cardiovasc. Res.* *119*, 857–866. <https://doi.org/10.1093/cvr/cvac099>.
32. Yang, B., Zhou, W., Jiao, J., Nielsen, J.B., Mathis, M.R., Heydarpour, M., Lettre, G., Folkersen, L., Prakash, S., Schurmann, C., et al. (2017). Protein-altering and regulatory genetic variants near GATA4 implicated in bicuspid aortic valve. *Nat. Commun.* *8*, 15481. <https://doi.org/10.1038/ncomms15481>.
33. Wang, S., Yu, H., Gao, J., Chen, J., He, P., Zhong, H., Tan, X., Staines, K.A., Macrae, V.E., Fu, X., et al. (2022). PALMD regulates aortic valve calcification via altered glycolysis and NF-κB-mediated inflammation. *J. Biol. Chem.* *298*, 101887. <https://doi.org/10.1016/j.jbc.2022.101887>.
34. Shelton, L., and Summers Rada, J.A. (2009). Inhibition of human scleral fibroblast cell attachment to collagen type I by TGFβ1p. *Investig. Ophthalmol. Vis. Sci.* *50*, 3542–3552. <https://doi.org/10.1167/iovs.09-3460>.
35. Rypdal, K.B., Apte, S.S., and Lunde, I.G. (2024). Emerging roles for the ADAMTS-like family of matricellular proteins in cardiovascular disease through regulation of the extracellular microenvironment. *Mol. Biol. Rep.* *51*, 280. <https://doi.org/10.1007/s11033-024-09255-5>.